

INNOVA JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
in preparation for General Certificate of Education Advanced Level  
**Higher 1**

CANDIDATE  
NAME

CLASS

INDEX NUMBER

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**MATHEMATICS**

**8865/01**

Paper 1

**24 August 2018**

**3 hours**

Additional Materials:      Answer Paper  
                                         Cover Page  
                                         List of Formulae (MF26)

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**READ THESE INSTRUCTIONS FIRST**

**Do not open this booklet until you are told to do so.**

Write your name, class and index number on the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use an HB pencil for any diagrams or graphs.  
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

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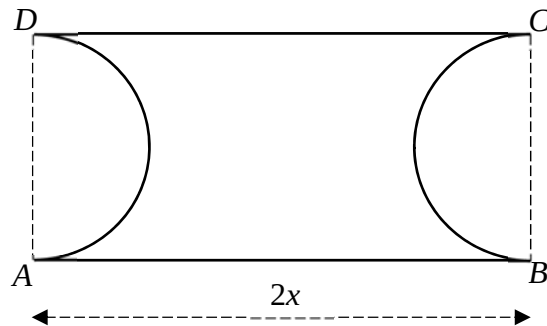
This document consists of **7** printed pages and **1** blank page.



**Section A: Pure Mathematics [40 marks]**

- 1 Find algebraically the set of values of  $k$  for which  
 $(k-2)x^2 - 2kx + (2k+3) < 0$   
 for all real values of  $x$ . [4]
- 2 (a) Differentiate  $\ln(\sqrt{1+3x^2})$  with respect to  $x$ . [2]
- (b) Find  $\int \frac{(2\sqrt{x}-1)^2}{\sqrt{x}} dx$ . [4]

3



A new design of a banner is created by cutting out two semicircles from the sides of a rectangular piece of cloth  $ABCD$  as shown in the diagram above. The length of  $AB$  is  $2x$  m and the total perimeter of the banner is 12 m.

- (i) Show that the area of the banner is  $\frac{12}{\pi}(3-x)(x-1)$  m<sup>2</sup>. [3]

The area of the banner is to be as large as possible.

- (ii) Use a non-calculator method to find the maximum value of this area, giving your answer in terms of  $\pi$ . Justify that this is the maximum value. [4]

- 4 The equation of a curve is  $y = \ln\left(3 - \frac{1}{2}x\right)$ .
- (i) Sketch the curve, stating the equations of any asymptotes and the exact coordinates of any points where the curve crosses the  $x$ - and  $y$ - axes. [3]
  - (ii) Find the equation of the tangent to the curve at the point where  $x = 2$ , giving your answer in the form  $px + qy = r$ , where  $p$  and  $q$  are integers and  $r$  is in an exact form. [4]
  - (iii) Find the numerical value of the area bounded by the curve, the tangent found in part (ii) and the  $x$ -axis. Leave your answer correct to 4 decimal places. [3]
- 5 Snape Industries produces three new chemical samples  $X$ ,  $Y$  and  $Z$ . Each sample is 200 millilitre (ml) and contains three types of compounds, namely, copper sulphate, sodium hydroxide and dextrose solution. The amount of copper sulphate, sodium hydroxide and dextrose solution in each sample is given below.

| Sample | Amount in ml for each 200 ml of sample |                  |                   |
|--------|----------------------------------------|------------------|-------------------|
|        | copper sulphate                        | sodium hydroxide | dextrose solution |
| $X$    | 80                                     | 60               | 60                |
| $Y$    | 120                                    | 60               | 20                |
| $Z$    | 60                                     | 100              | 40                |

One of the researchers, Herman, wants to experiment producing a new chemical sample  $W$  of volume 200 ml by mixing different amounts of  $X$ ,  $Y$  and  $Z$  such that there are 75 ml of copper sulphate, 75 ml of sodium hydroxide and 50 ml of dextrose solution.

- (i) Determine the amount, in ml, of samples  $X$ ,  $Y$  and  $Z$  in this sample of  $W$ . [4]

Herman then conducts a further experiment on sample  $W$ . He finds that the potential energy,  $P$  kJ, of the chemical compounds present after  $t$  seconds can be modelled by the equation

$$P = 8 + 5e^{-0.4t} - e^{1-2t}.$$

- (ii) Sketch the graph of  $P$  against  $t$  for  $t \geq 0$ , stating the equations of any asymptotes and the coordinates of the intersection with the  $P$ -axis. [3]
- (iii) Use your calculator to estimate the maximum value of  $P$ . State also the value of  $t$  for which this maximum value occurs. [2]
- (iv) Find the rate at which the potential energy is changing when  $t = 3$ . [1]
- (v) Find the exact area of the region bounded by the curve, the line  $t = 0.5$  and both axes. [3]

**Section B: Statistics [60 marks]**

- 6** A confectionery produces cups of jelly for sale. The masses of these cups of jelly have a normal distribution. Over a long period of time, it is found that 3% of them have a mass less than 40 grams and 8% of them have a mass more than 50 grams. Find the mean and variance of this distribution. [4]

- 7** An international spy, Johnny Bond, is tasked to form 8-letter codes using the letters of the word QUESTION. Find the number of codes that he can form if

- (i) all the vowels are next to each other, [2]
- (ii) the first and the last letters are consonants. [2]

Johnny Bond believes that a code is ‘good’ if the consonants and vowels alternate.

- (iii) Find the probability of forming a ‘good’ code. [3]

- 8** At a carnival, Abel plays a game in which he is given two tries of throwing balls at a row of cans with the aim of toppling any one can. The probability of a can containing a prize is 0.3. The probability that he successfully topples any can in his first try is 0.25. If Abel fails to do so in his first try, he will be given a second try and the probability that he will successfully topple any can increases to 0.4. The game ends when he successfully topples a can or he misses on his second try.

- (i) Draw a tree diagram to represent the possible outcomes. [2]
- (ii) Find the probability that Abel does not win any prize. [2]
- (iii) Given that Abel wins a prize, find the probability that it is from his second try. [2]

Abel plays the game five times.

- (iv) Find the probability that Abel wins a prize in at least 3 out of the 5 games. [3]

- 9** A company produces a large number of graphing calculators and on average, 5% of the graphing calculators are faulty. The graphing calculators are being packed into boxes of 30.

- (i) Explain the significance of the phrase ‘large number’ in the first sentence of the question. [1]
- (ii) Find the probability that there are more than 3 faulty graphing calculators in a randomly chosen box. [2]
- (iii) 10 boxes are randomly selected. Find the probability that the 10<sup>th</sup> box is the second box that contains more than 3 faulty graphing calculators. [3]
- (iv) 50 boxes of 30 graphing calculators each are randomly chosen. Find the probability that the mean number of faulty graphing calculators in a box is less than 1.6. [3]

- 10** A birdwatcher claims that the mean wing length of English male blackbirds is 131 mm . An ornithologist doubts this claim and wishes to investigate if this claim is valid by carrying out a hypothesis test. A random sample of 40 English male blackbirds is obtained and the wing length of each bird is recorded. The mean and standard deviation of this sample are found to be 127.5 mm and 11.84 mm respectively.

- (i) Explain whether the ornithologist should use a 1-tail test or a 2-tail test. [1]
- (ii) Explain why the ornithologist is able to carry out a hypothesis test without knowing anything about the distribution of the wing lengths of English male blackbirds. [1]
- (iii) Using your answer to part (i), find the  $p$ -value of the test and state the meaning of this  $p$ -value in context. [3]

The birdwatcher now claims that the mean wing length of English male blackbirds is at most 131 mm. The ornithologist then takes a second random sample of 40 such birds and record their wing lengths. The unbiased estimates for the population mean and variance are  $k$  mm and 150.7984 mm<sup>2</sup> respectively. Using a test at 5% significance level, the ornithologist finds that the birdwatcher’s claim is invalid.

- (iv) Find the set of values within which  $k$  must lie. Leave your answer correct to 1 decimal place. [4]

- 11** The total number of new subscribers to an online shopping site,  $x$ , as well as the total amount of profit generated,  $y$  thousand dollars, from these subscribers are recorded for 8 randomly chosen weekends. The table below gives a summary of the results.

|     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $x$ | 590 | 900 | 850 | 680 | 700 | 790 | 630 | 880 |
| $y$ | 89  | 133 | 125 | 106 | 99  | 123 | 102 | 139 |

- (i)** Give a sketch of the scatter diagram for this set of data. [2]
- (ii)** Find the product moment correlation coefficient and comment on its value in the context of the data. [2]
- (iii)** Find the equation of the regression line of  $y$  on  $x$  in the form of  $y = mx + c$ , giving the values of  $m$  and  $c$  correct to 3 decimal places. Sketch this line on your scatter diagram. [2]
- (iv)** Give interpretations of the meaning of the two values  $m$  and  $c$ , in the context of the data. [2]
- (v)** Using the equation of your regression line, estimate the number of new subscribers to this online shopping site for a weekend in which the profit amount of \$100,000 is collected. Comment on the reliability of this estimate. [2]

- 12** Potatoes and carrots are sold by weight in a market. The masses, in kilograms (kg), of potatoes and carrots are modelled as having independent normal distributions with means and standard deviations as shown in the table.

|          | Mean  | Standard Deviation |
|----------|-------|--------------------|
| Potatoes | 0.165 | 0.015              |
| Carrots  | 0.072 | 0.012              |

- (i)** Stating clearly the mean and variance of the distribution that you use, find the probability that the total mass of 2 randomly chosen carrots is within 20 g of the mass of a randomly chosen potato. [4]

Potatoes cost \$2.50 per kilogram and carrots cost \$1.80 per kilogram.

- (ii)** Two potatoes are randomly chosen. Find the probability that each of them cost more than \$0.40. [3]
- (iii)** Without any calculation, explain why the probability that the total cost of two randomly chosen potatoes is more than \$0.80 will be greater than your answer to part **(ii)**. [1]

Mathew buys 8 potatoes and 6 carrots as part of his ingredients to cook a vegetable stew.

- (iv)** Stating clearly the mean and variance of the distribution that you use, find the probability that Matthew spends between \$3.80 and \$4.20. [4]

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